

HolisticX Cloud Architecture

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1 Overall Architecture Review

The HolisticX architecture is a modern microservices-based system built for scalability, resilience, and security on AWS. It follows a layered design consisting of a secure network layer, a container-based application layer, a managed data layer, and a robust edge layer for handling incoming requests.

The most significant architectural feature is the **complete isolation of the Keycloak identity provider**. Keycloak operates in its own dedicated network stack with its own load balancer, compute cluster, and database, ensuring that the critical authentication service is fully decoupled from the main application stack. This enhances security, stability, and independent scalability.

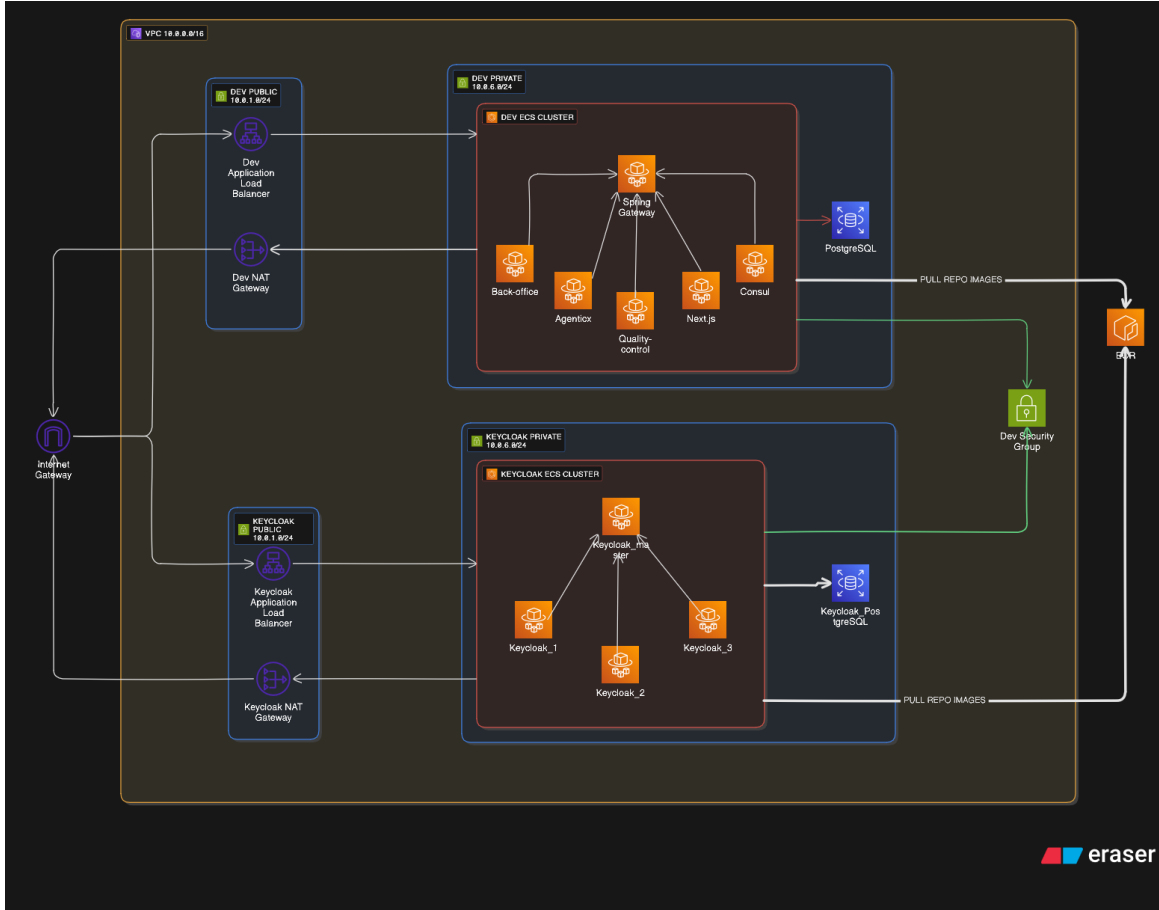


Figure 1: High-Level Architecture Overview

2 Network Layer: The Foundation

This layer, primarily defined in `modules/network/main.tf`, creates a secure and isolated environment within an AWS Virtual Private Cloud (VPC) (10.0.0.0/16). The VPC is logically partitioned into two distinct network stacks: one for the main application and one for Keycloak.

2.1 Main Application Network Stack

- **Public Subnets (e.g., 10.0.1.0/24):** Host the main Application Load Balancer (ALB) and the shared NAT Gateway.
- **Private Subnets (e.g., 10.0.3.0/24):** Host the core application microservices on ECS and the shared application RDS database.

2.2 Keycloak Network Stack (Isolated)

- **Keycloak Public Subnets (e.g., 10.0.5.0/24):** Host the dedicated Keycloak Application Load Balancer.
- **Keycloak Private Subnets (e.g., 10.0.6.0/24):** Host the Keycloak ECS cluster and its dedicated RDS database.

An **Internet Gateway (IGW)** provides internet access for the VPC, while a **NAT Gateway** allows private resources to initiate outbound connections without being publicly accessible.

3 Application and Compute Layer

This layer runs on AWS ECS with Fargate, a serverless compute engine that removes the need to manage underlying EC2 instances.

- **Main Application Cluster:** Hosts the core microservices:
 - **spring-gateway:** The central API Gateway and router.
 - **nextjs:** The frontend user interface.
 - **agentix, backoffice:** Core backend services (more will be added).
 - **consul:** Provides service discovery for the backend services.
- **Keycloak Cluster:** A dedicated cluster running two Keycloak instances for high availability.

All services use Docker images stored in dedicated **AWS ECR repositories** and are configured to run on cost-effective **Graviton (ARM64)** processors (20% cheaper than the x86-based instances).

4 Data and Observability Layer

- **Application RDS:** A PostgreSQL database instance shared by the main application microservices.
- **Keycloak RDS:** A dedicated PostgreSQL database instance used exclusively by Keycloak.
- **Secrets Management:** All database passwords are created, stored, and rotated automatically by **AWS Secrets Manager**. Applications retrieve credentials dynamically at runtime via their IAM role.
- **Observability:** **AWS X-Ray** provides distributed tracing, and **AWS CloudWatch** collects logs from all services with a 30-day retention policy.

5 Security: IAM Roles Security Groups

Security is enforced through a combination of network-level and identity-level controls.

5.1 Security Groups (Virtual Firewalls)

- **Main ALB SG:** Allows public HTTPS (443) traffic.
- **Main ECS SG:** Allows traffic from the Main ALB SG on application ports.
- **Main RDS SG:** Allows PostgreSQL (5432) traffic only from the Main ECS SG.
- **Keycloak ALB SG:** Allows public HTTPS (443) traffic.
- **Keycloak ECS SG:** Allows traffic from the Keycloak ALB SG on port 8080.
- **Keycloak RDS SG:** Allows PostgreSQL (5432) traffic only from the Keycloak ECS SG.

5.2 IAM Roles (Permissions)

- **Task Execution Role:** Grants the ECS service permissions to pull images from ECR and send logs to CloudWatch.

Key Permissions:

- `ecr:GetAuthorizationToken`,
`ecr:BatchCheckLayerAvailability`,
`ecr:GetDownloadUrlForLayer`,
`ecr:BatchGetImage`:
Allows ECS to pull your container images from your ECR repositories.
- `logs:CreateLogStream`, `logs:PutLogEvents`:
Allows ECS to send your container logs to AWS CloudWatch.

- **Task Role:** Grants the application code permissions to access other AWS services, such as retrieving passwords from Secrets Manager or sending traces to X-Ray.

Key Permissions:

- `secretsmanager:GetSecretValue`:
This is the crucial permission that allows your application to ask Secrets Manager for the database password.
- `s3:PutObject`, `s3:GetObject`:
Grants permission to read from and write to the RDS backup S3 bucket.
- `xray:PutTraceSegments`:
Allows your application to send tracing data to AWS X-Ray for performance monitoring.

6 DNS and Request Flow

6.1 DNS Configuration (Route 53)

A single **Route 53 Hosted Zone** for `holisticx.com` manages all DNS records. This requires a one-time manual update at the domain registrar (IONOS) to delegate authority to the AWS name servers. Terraform automatically creates two primary records:

- `dev.holisticx.com` points to the Main Application ALB.
- `keycloak.holisticx.com` points to the dedicated Keycloak ALB.

7 Conclusion

This revised architecture provides a robust, scalable, and highly secure foundation for the HolisticX platform. The complete isolation of the Keycloak identity service is a critical security enhancement that aligns with modern best practices for cloud-native applications.